

COMPONENTS

COMPONENT	SPECIFICATION	FUNCTION
ESP32	Dual-core Wi-Fi & Bluetooth MCU	Main controller of the system
DHT22 Sensor	Temperature & Humidity Sensor	Monitors environmental conditions
MQ-135 Sensor	Gas sensor	Detects harmful gases and smoke
PIR Sensor	Passive Infrared Sensor	Detects human motion
OLED Display	0.96-inch I2C OLED Display	Displays system status and sensor readings
Relay Module	5v Relay Module	Controls connected appliances
Buzzer	5v Active Buzzer	Provides audible alerts
Boost Converter	DC-DC Step-Up Converter	Provides stable voltage supply where required
Power Supply Unit	5v DC Supply	Powers the system
Connecting Wires	Jumper Wires	Facilitates component connections